

- 5 (a) e.g. separate charges, store energy, smoothing circuit. etc. B1 [1]
 (allow 'stores charge')
- (b) (i) charge = current $\times$ time B1 [1]
- (ii) area is 21.2 cm^2 (allow $\pm 0.5 \text{ cm}^2$) C2
 (allow 1 mark if outside $\pm 0.5 \text{ cm}^2$ but within $\pm 1.0 \text{ cm}^2$)
 1.0 cm^2 represents $(0.125 \times 10^{-3} \times 1.25 =) 156 \mu\text{C}$ C1
 charge = $3300 \mu\text{C}$ A1 [4]
- (iii) capacitance = Q/V C1
 $= (3300 \times 10^{-6}) / 15$
 $= 220 \mu\text{F}$ A1 [2]
- (c) either energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
 $\frac{1}{2} \times C \times 15^2 = 2 \times \frac{1}{2} \times C \times V^2$ C1
 $V = 10.6 \text{ V}$ A1 [3]
- 6 (a) (i) $BI \sin \theta$ B1 [1]